The Used Cars Investigation

A Data-Driven Storytelling Adventure

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Abstract

This project was developed by students of **AI/Data Science** at **ENSAM RABAT**, under the valuable supervision of **Professor M. El Yadari**. We thank him for the valuable insights we learned from him and we hope to learn more from him in the future.

The Mission Begins…

> *Intuition is thinking that you know without knowing why you do. -Daniel Kahneman*

Our mission — should we choose to accept it — is to:

1. **Clean** this wild, messy dataset
2. **Explore** its hidden patterns
3. **Build** a machine learning model to predict used car prices

Let's follow the data.

The Dataset at a Glance

Our dataset has **15 columns** — each one a clue:

> *Data Source: We got this dataset from Mendeley. On a side note, we thank the people that kindly curated this data for us to work with.*

Column	What it tells us
`Brand`, `Model`	Which car it is
`Year`, `Mileage`	How old / used it is
`Price`	What we want to predict

Column	What it tells us
`Condition`, `Gearbox`, `Fuel`	Car characteristics
`Location`, `Origin`	Where it came from
`First Owner`, `Equipment`	Status & extras
`Fiscal Power`, `Sector`, `Number of Doors`	Technical details

▶ Code

```
Dataset: 101,896 rows × 15 columns
```

Part 1: The data

Dealing with Missing Data

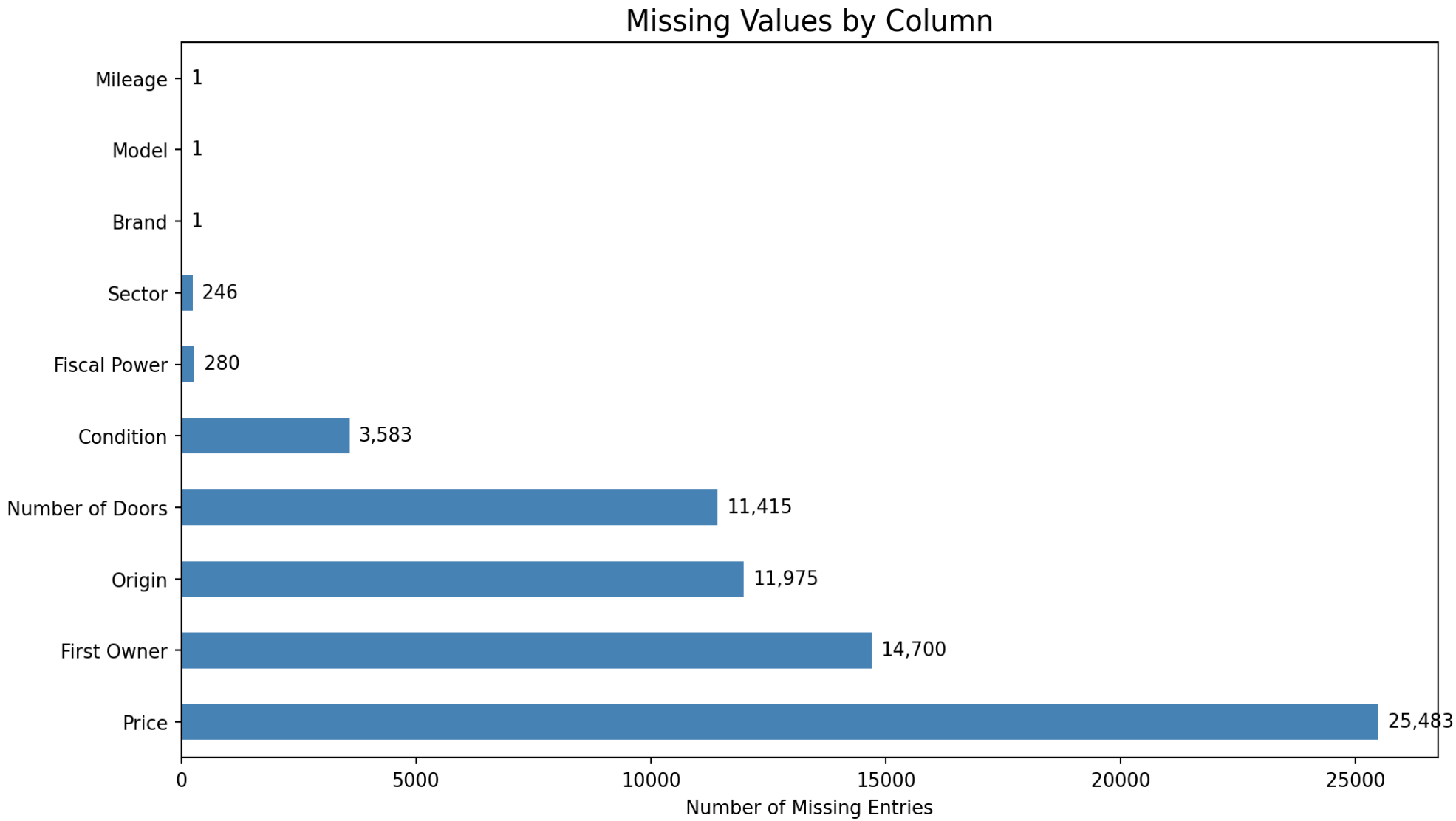
The first thing any good data scientist does — **investigate the data**.

▶ Code

```
About 4.43% of the data is missing
Brand                  1
Model                  1
Condition           3583
Mileage                1
Fiscal Power         280
Number of Doors    11415
Origin             11975
First Owner        14700
Sector               246
Price              25483
dtype: int64
```

The Missing Data Map

- Code



Several columns have missing values. Let’s tackle them one by one!

Q: What do we do with cars that have no Price?

> **Answer:** Drop them! `Price` is our target variable — without it, a record is useless.

▶ Code

```
Dropped 25483 rows with missing Price
Remaining rows: 76,413
```

Q: What about the Number of Doors?

> **Answer:** We can **infer** it! Cars of the same Brand, Model, and Year almost always share the same door count.

▶ Code

```
Missing 'Number of Doors' after imputation: 0
```

Strategy: Most specific group first (Brand + Model + Year), then fall back gracefully.

Q: What about the other columns?

Column	Strategy	Why
`Condition`	**Drop** rows	Can't reliably infer condition
`Origin`	**Fill** → `"Unknown"`	Better than losing the row
`First Owner`	**Fill** → `"Unknown"`	Could be any owner
`Fiscal Power`	**Impute** by Brand+Model+Year	Same car = same engine

Column	Strategy	Why
`Sector`	**Fill** → `"Unknown"`	Optional metadata
`Brand / Model`	**Drop** the one bad row (#6714)	No way to recover

▶ Code

```
Clean dataset shape: (72239, 15)
```

Q: What about the Location column?

> **Problem:** Some locations are in French whilst others are in Arabic. **Answer:** We need to make this data uniform! Since the majority of the city names are in French, we decided to convert all the Arabic city names to their corresponding French names.

- Code

```
Total unique locations after mapping: 364
```

Q: Should we rename columns?

> **Problem:** Some columns have long names and are cumbersome to work with. **Answer:** To make the dataset easier to work with, we could rename columns for easier column selection and more intiutive names by just taking the initials

- Code

Q: What about fiscal power?

> **Answer:** We can **infer** it! Cars of the same Brand, Model, and Year almost always share the same fiscal power.

- Code

```
Missing 'Fiscal Power' after imputation: 0
```

The Dataset is Now Clean!

▶ Code

```
Zero missing values remaining!
<class 'pandas.core.frame.DataFrame'>
Index: 72239 entries, 0 to 101892
Data columns (total 15 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Brand         72239 non-null  object
 1   Model         72239 non-null  object
 2   Year          72239 non-null  int64
 3   Condition     72239 non-null  object
 4   Mileage       72239 non-null  object
 5   Gearbox       72239 non-null  object
 6   Fiscal Power  72239 non-null  object
 7   Fuel          72239 non-null  object
```

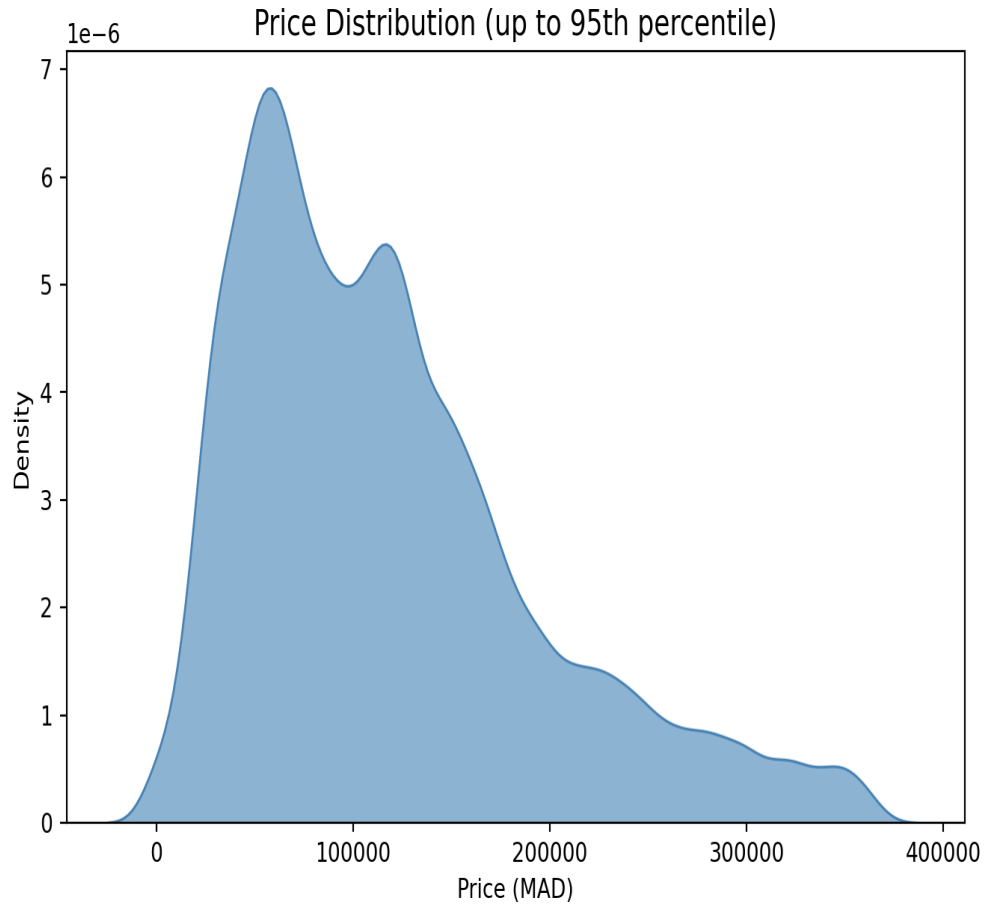
> *The crime scene has been processed. Now comes the real*

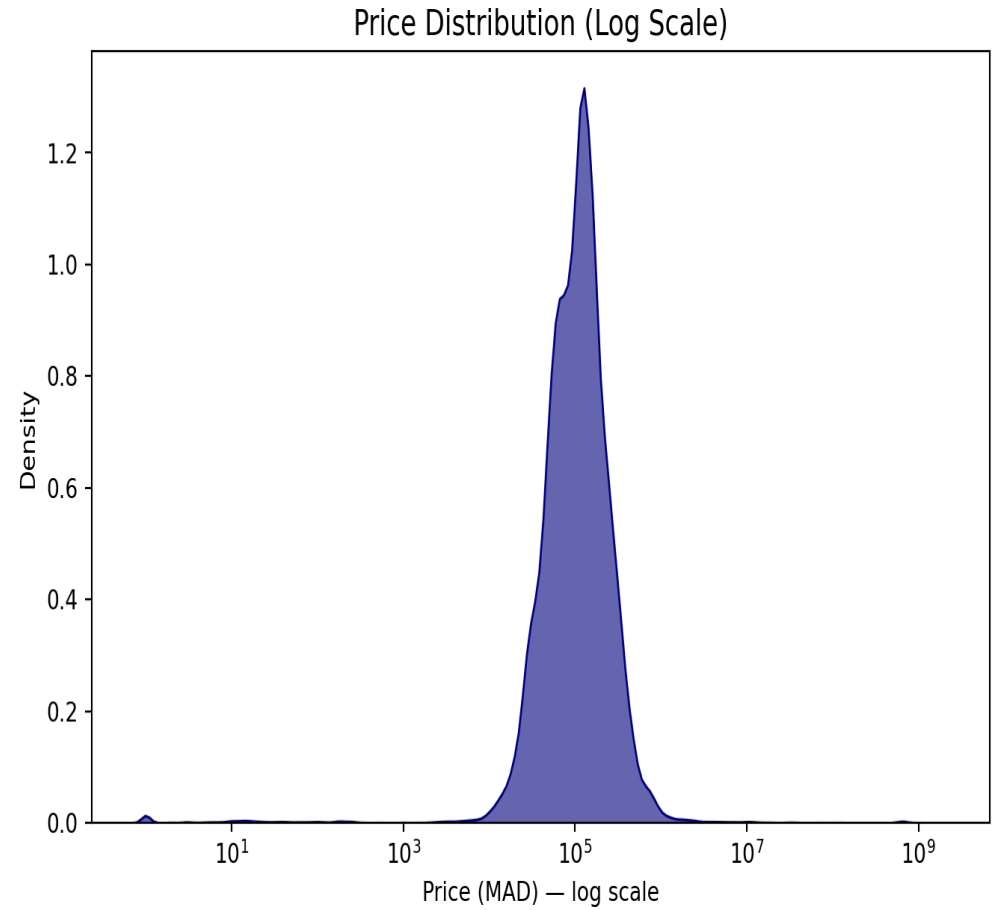
Part 2: The Price Enigma

Q1: What does the price distribution look like?

Is it normal? Skewed? Could it be… hiding outliers?

▶ Code

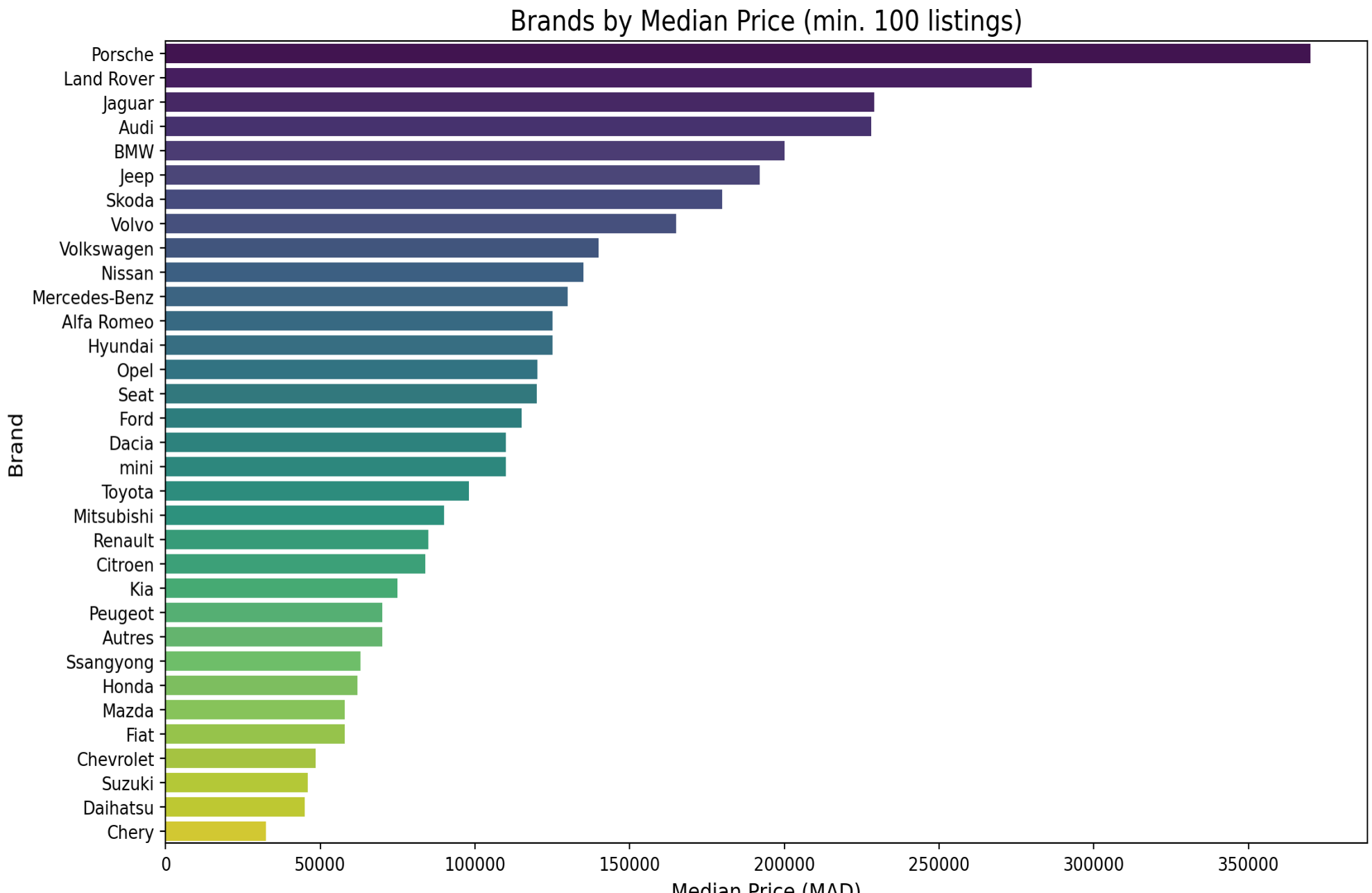




> **Answer:** The distribution is **heavily left-skewed** — most cars cluster in the affordable range, but luxury outliers shoot the mean upward. Log scale reveals a more symmetric shape.

Q2: Which brands command the highest prices?

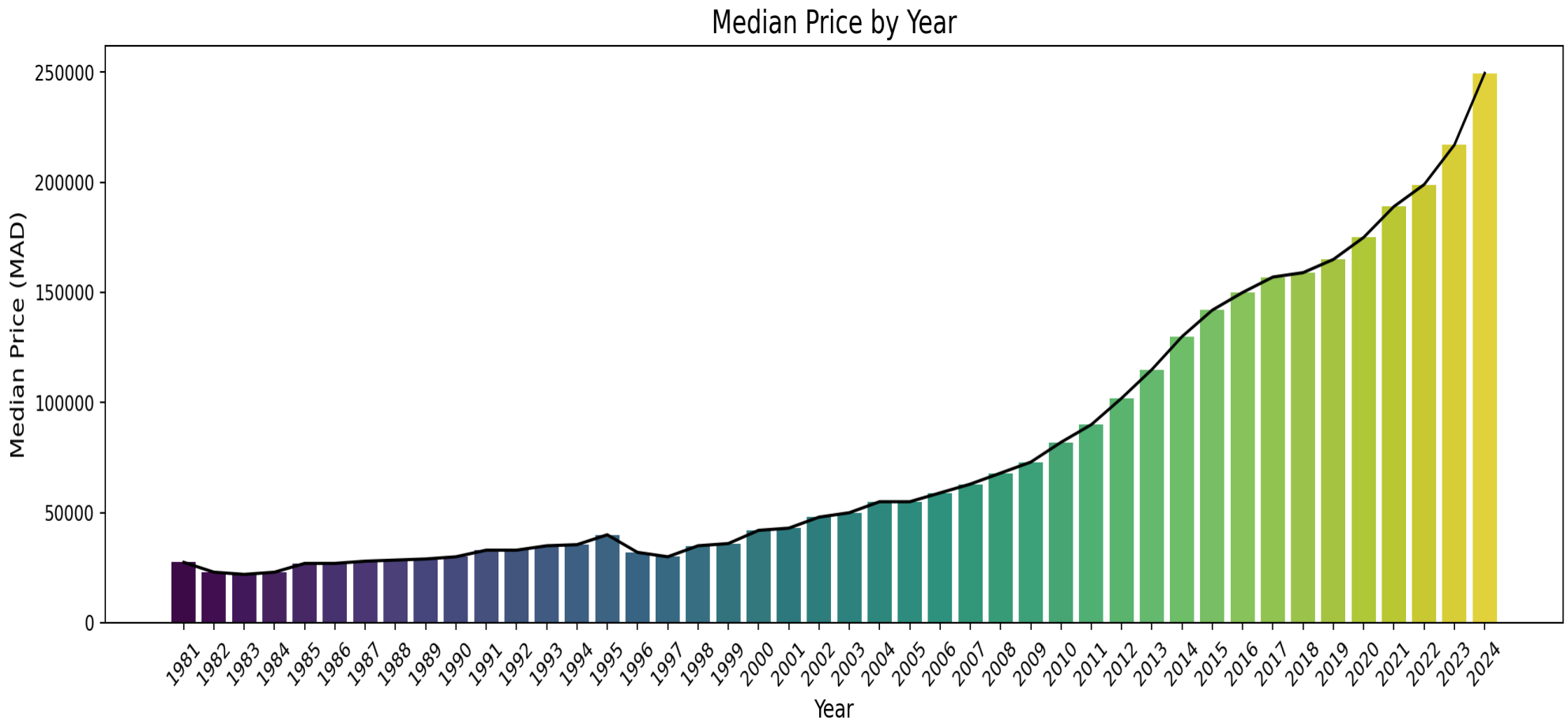
▶ Code


Brands by Median Price (min. 100 listings)
Porsche
Land Rover
Jaguar
Audi
BMW
Jeep
Skoda
Volvo
Volkswagen
Nissan
Mercedes-Benz
Alfa Romeo
Hyundai
Opel
Seat
Ford
Dacia
mini
Toyota
Mitsubishi
Renault
Citroen
Kia
Peugeot
Autres
Ssangyong
Honda
Mazda
Fiat
Chevrolet
Suzuki
Daihatsu
Chery
Brand
0
50000
100000
150000
200000
250000
300000
350000
Median Price (MAD)

Q3: Does the year affect the price?

> *Intuitively: older car = lower price. But is that always true?*

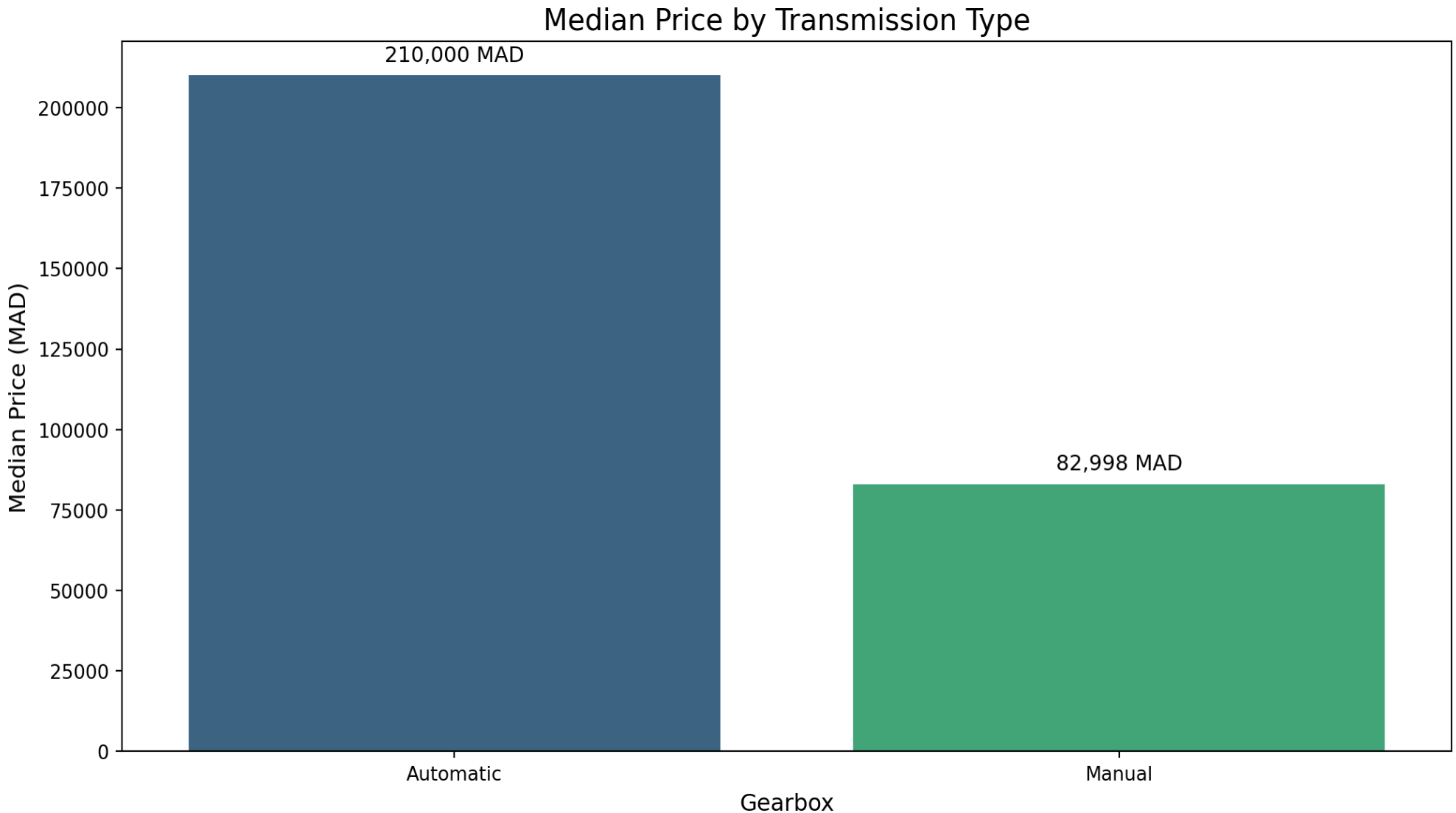
- Code



> **Answer:** Yes! The trend is clear — **newer cars command higher prices**, with a strong upward slope from 2010 onward.

Q4: Manual vs. Automatic — Who pays more?

- Code

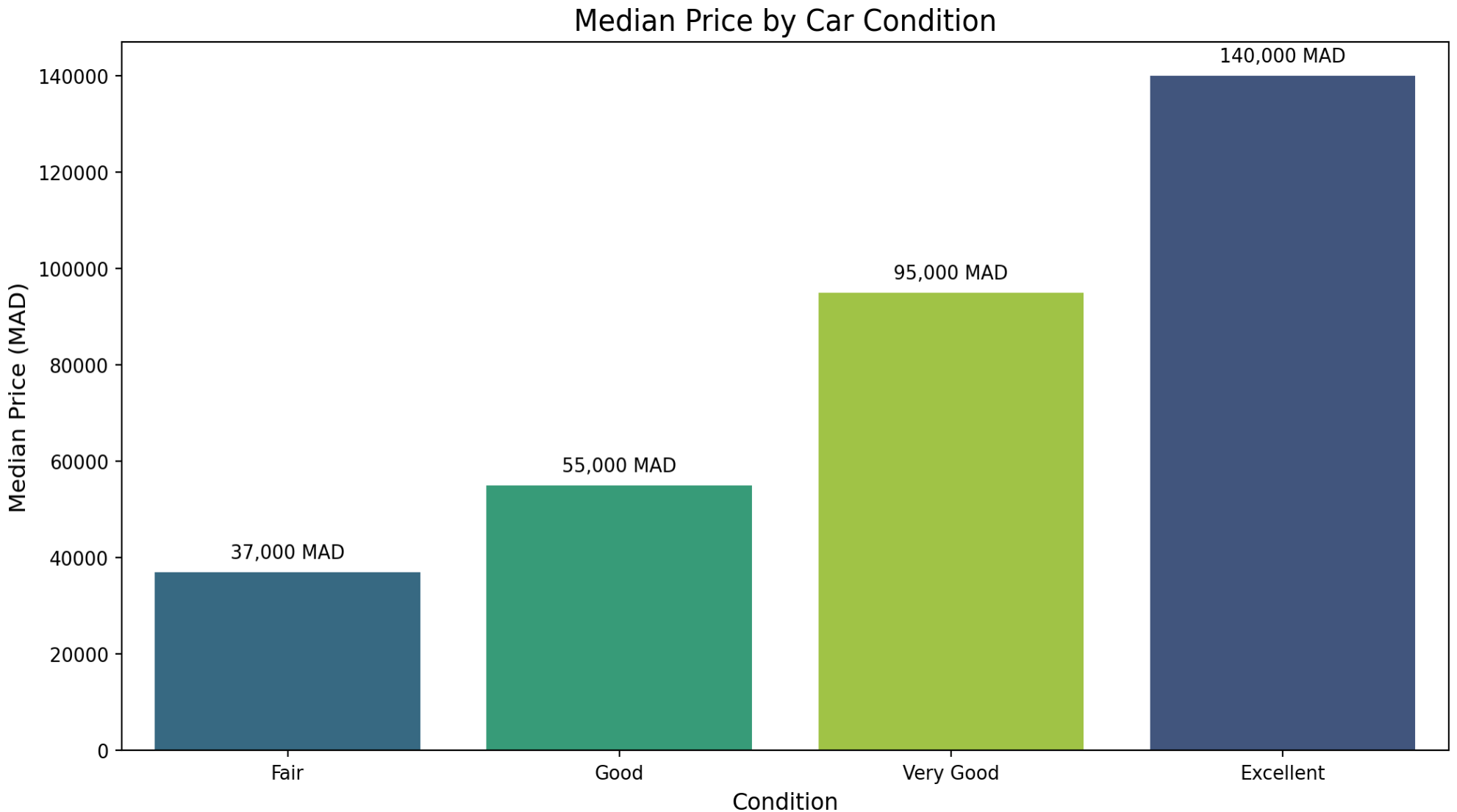


> **Answer: Automatic cars are significantly pricier** — they're often linked to newer, higher-end models.

Q5: Does Condition match our expectations?

> *We expect: Fair < Good < Very Good < Excellent*

- Code

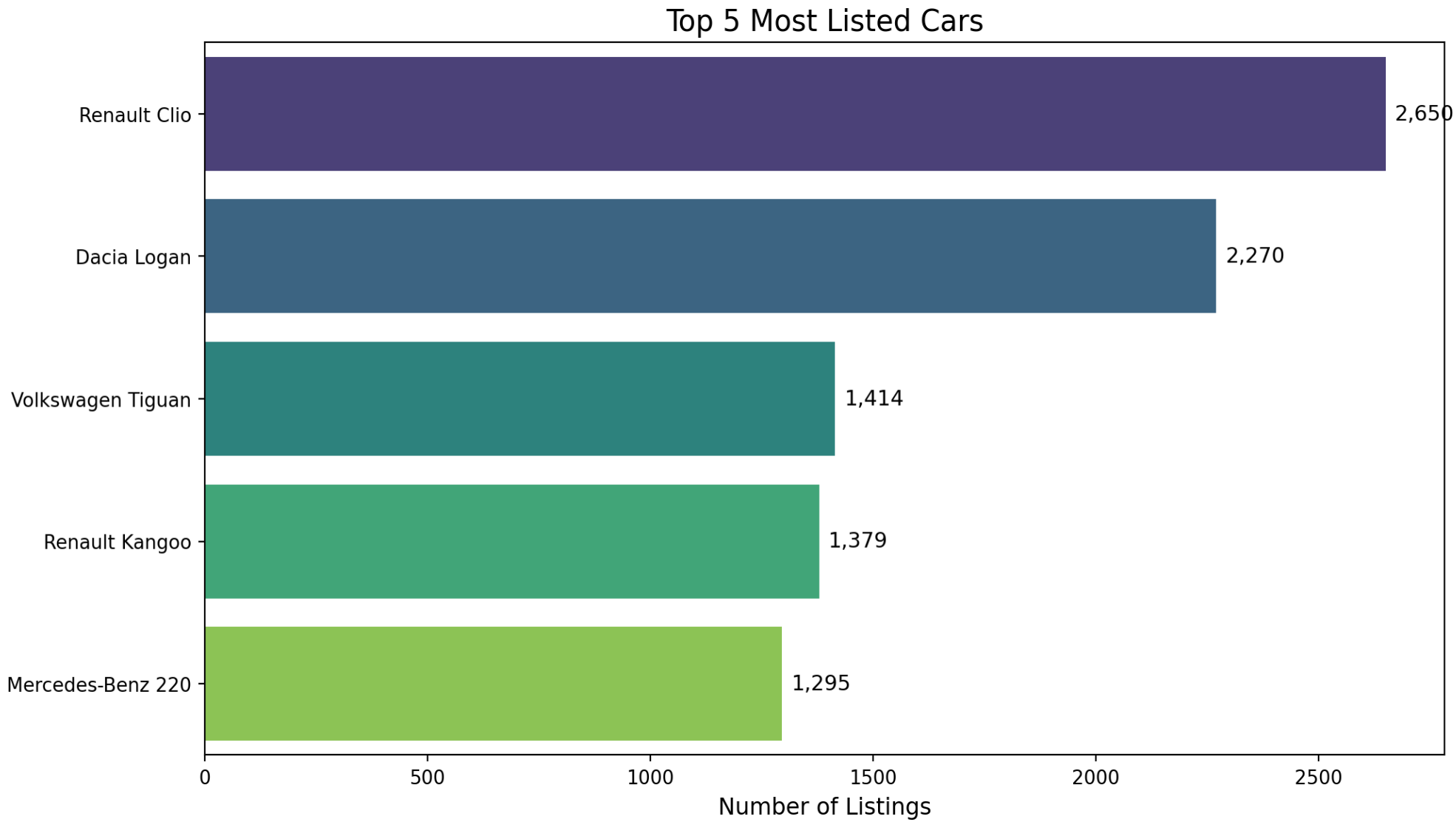


Answer: Expectations confirmed! Excellent condition cars command the highest price.

Part 3: The Market Map

Q6: Who dominates the market?

- Code

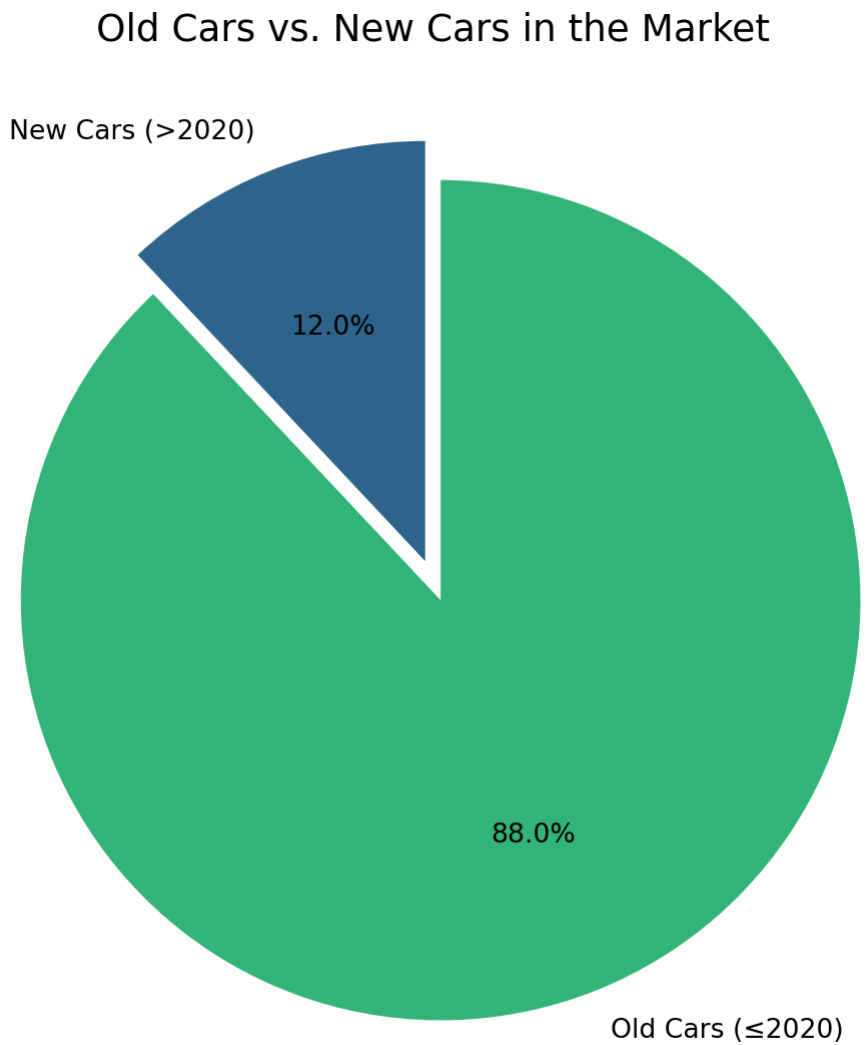


Answer: The **Renault Clio and Dacia Logan** lead the pack — affordable, reliable, and everywhere!

Q7: Old cars or New cars?

> *We consider New = After 2020. Old = 2020 and before.*

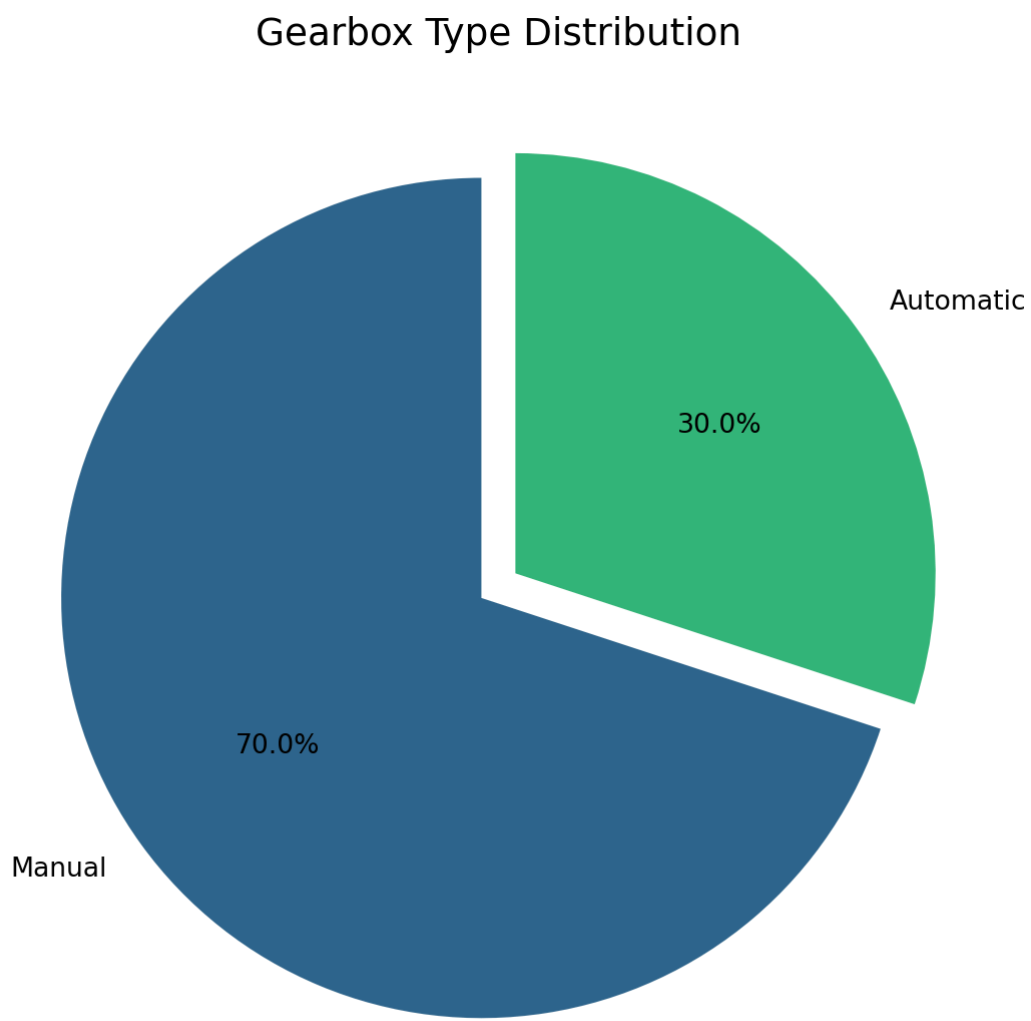
- Code



Answer: The market is dominated by **older cars** — the used car market is truly the domain of pre-2020 vehicles.

Q8: Manual or Automatic?

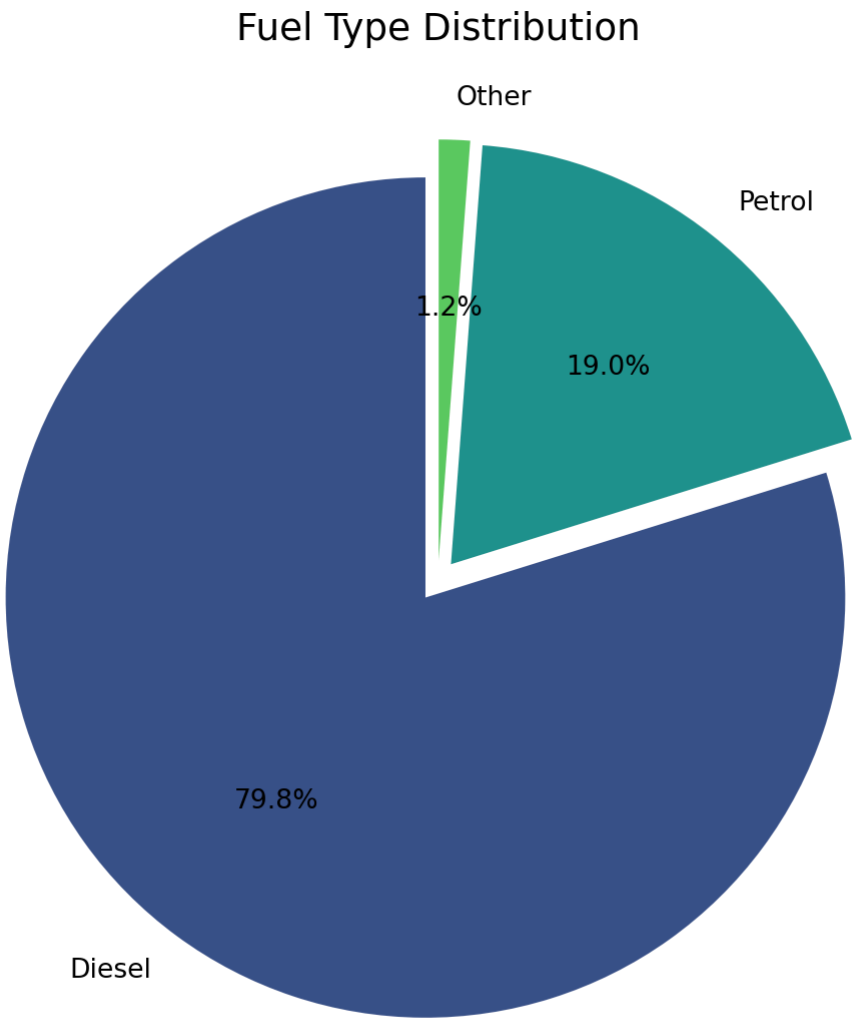
- Code



Answer: Manual transmissions **dominate** the used car market in Morocco — automatics remain a premium niche.

Q9: Diesel or Petrol?

- Code

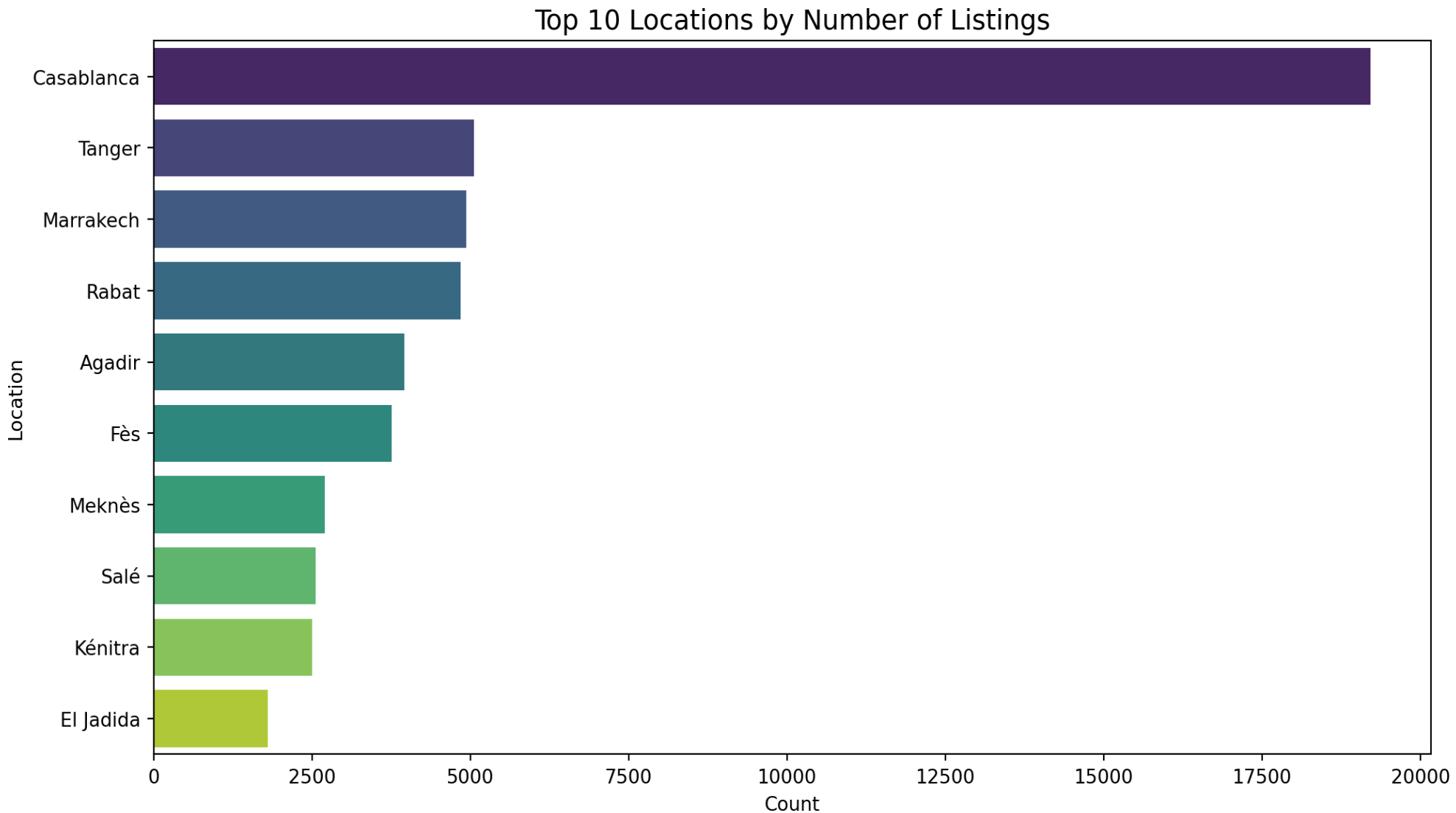


Answer: Diesel leads the way — a reflection of Moroccan driving culture where diesel is preferred for long distances

Part 4: The Geographic Trail

Q10: Where are all these cars coming from?

- Code



Answer: Casablanca massively leads in listings — Morocco's commercial capital is also its used car capital.

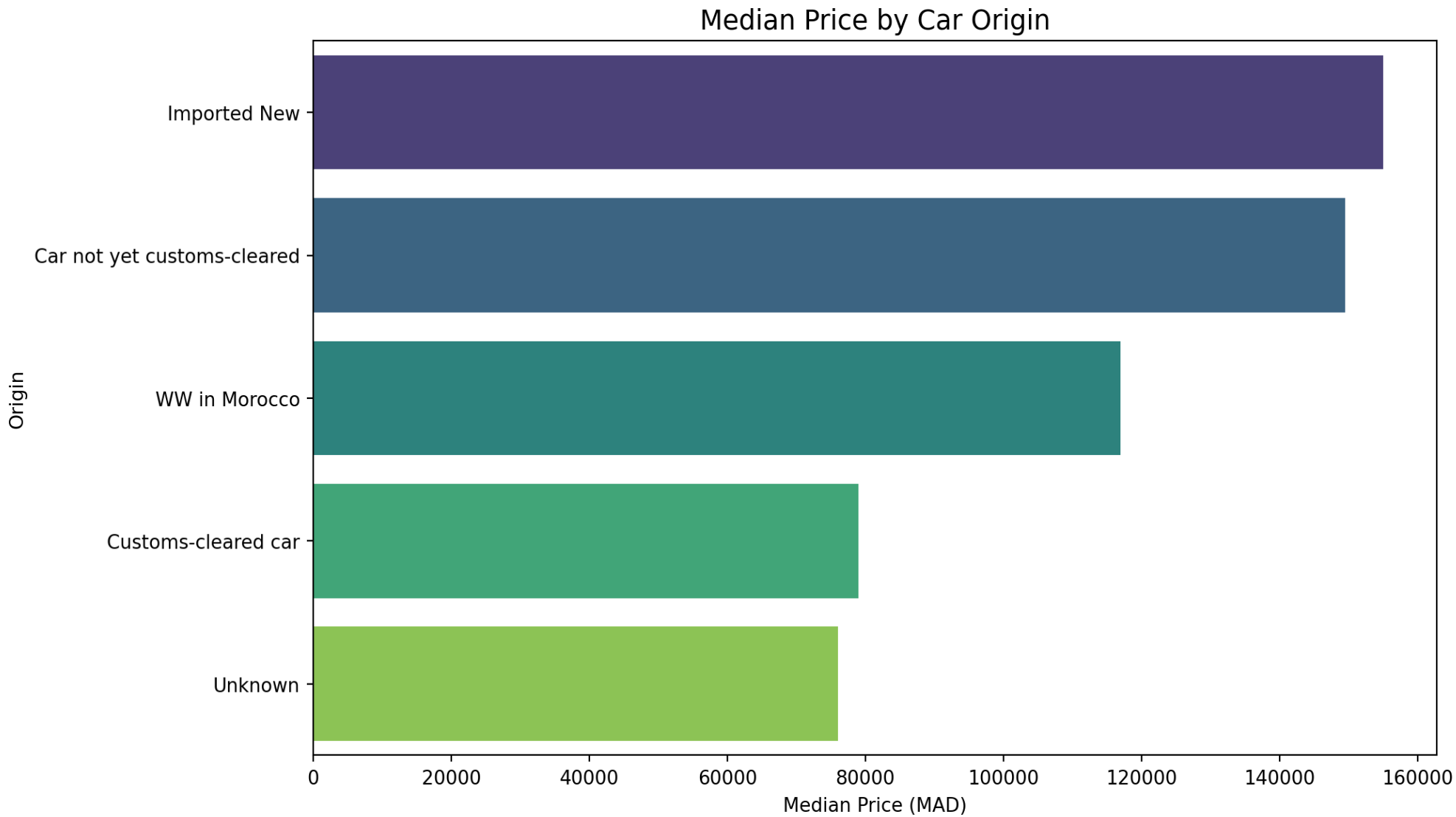
Q11: Where are the most expensive cars?

- Code

Answer: *Location matters!* Some cities command significantly higher median prices — likely reflecting

Q12: Does the Origin affect price?

- Code

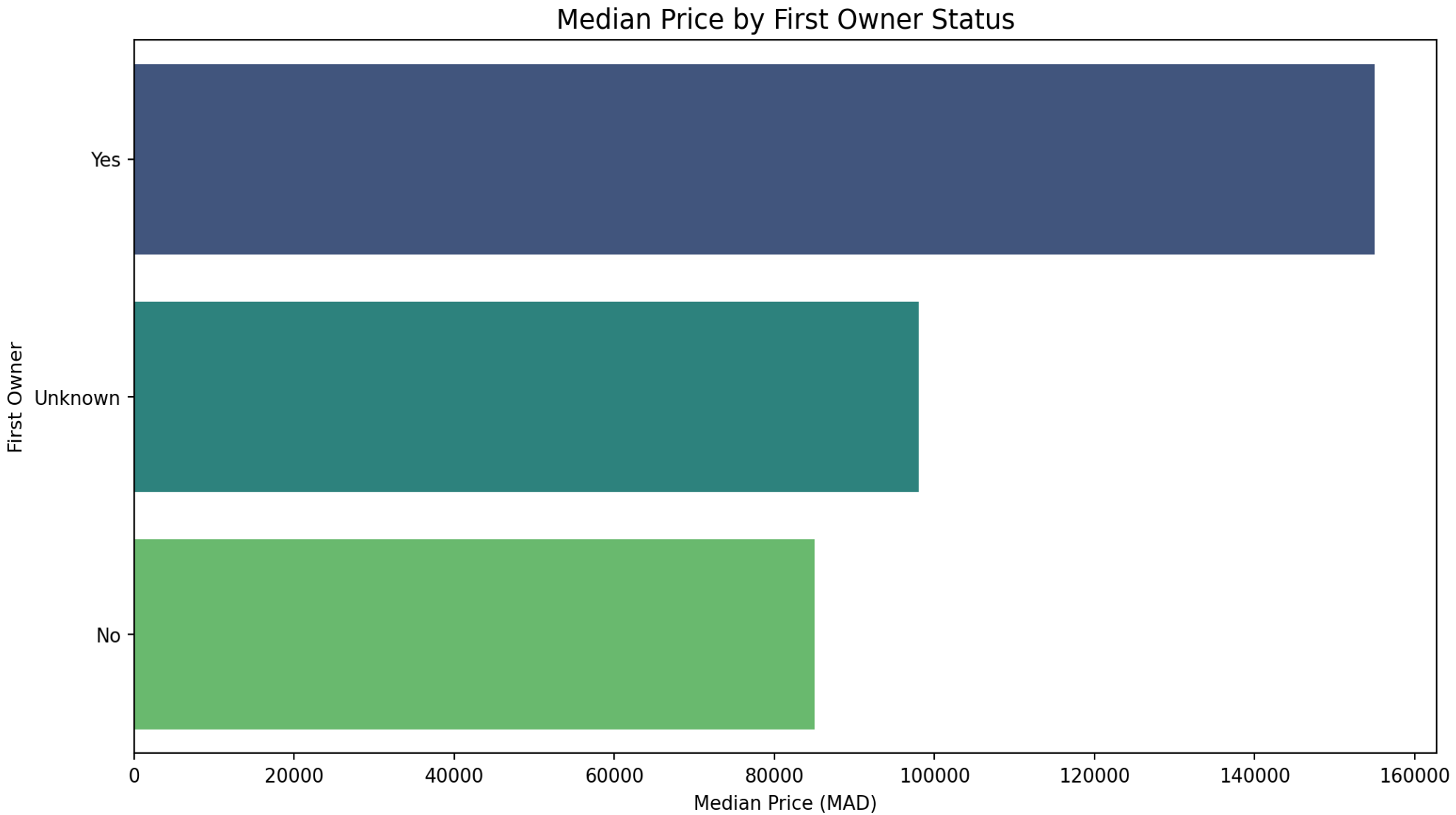


Answer: Yes! **Imported and customs-cleared** cars tend to command a premium over locally-registered vehicles.

Part 5: The Hidden Factors

Q13: Does being First Owner matter?

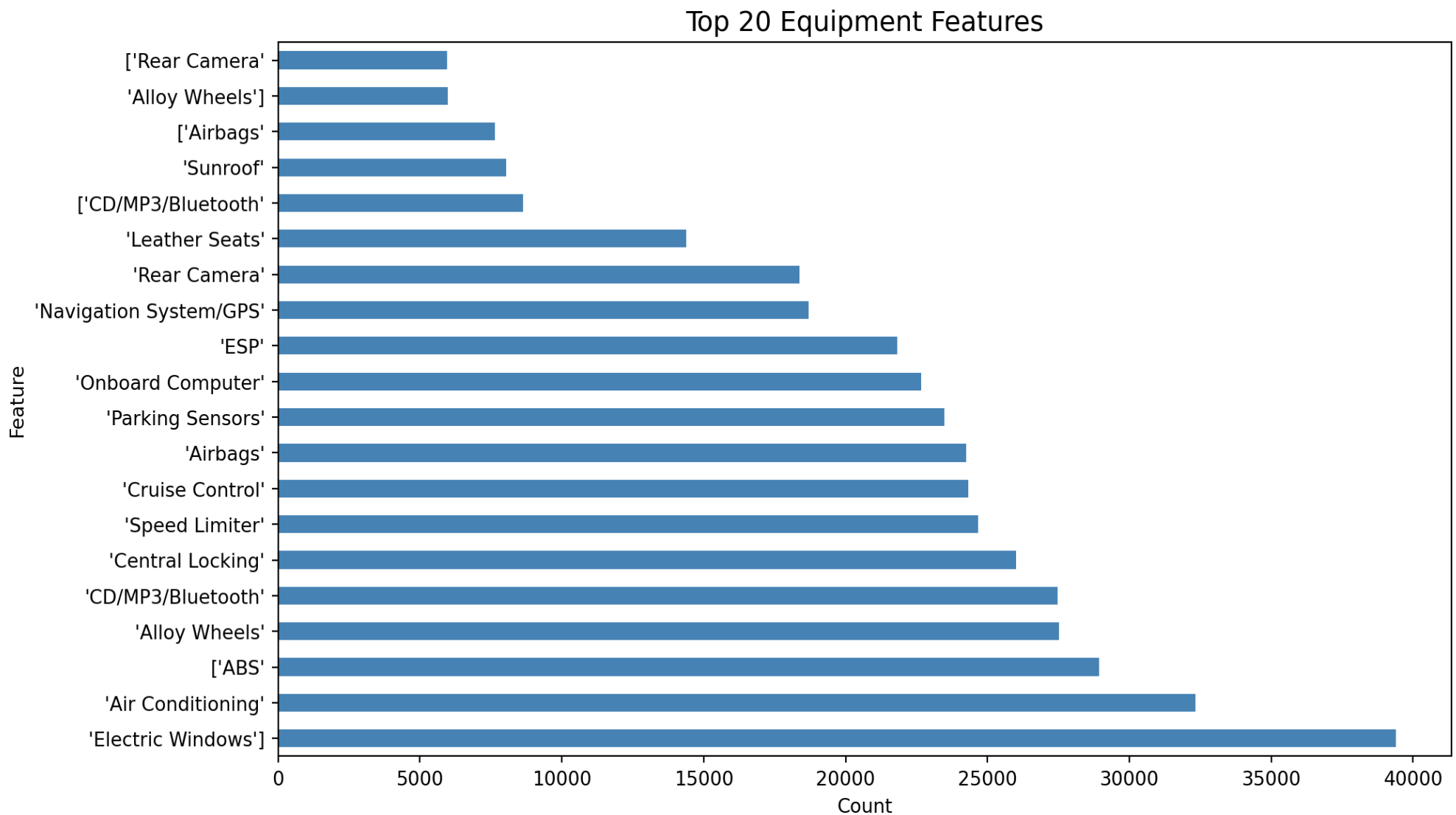
- Code



Answer: Yes! First-owner cars clearly command a premium. Buyers trust a single-owner history.

Q14: What equipment features are most common?

▶ Code

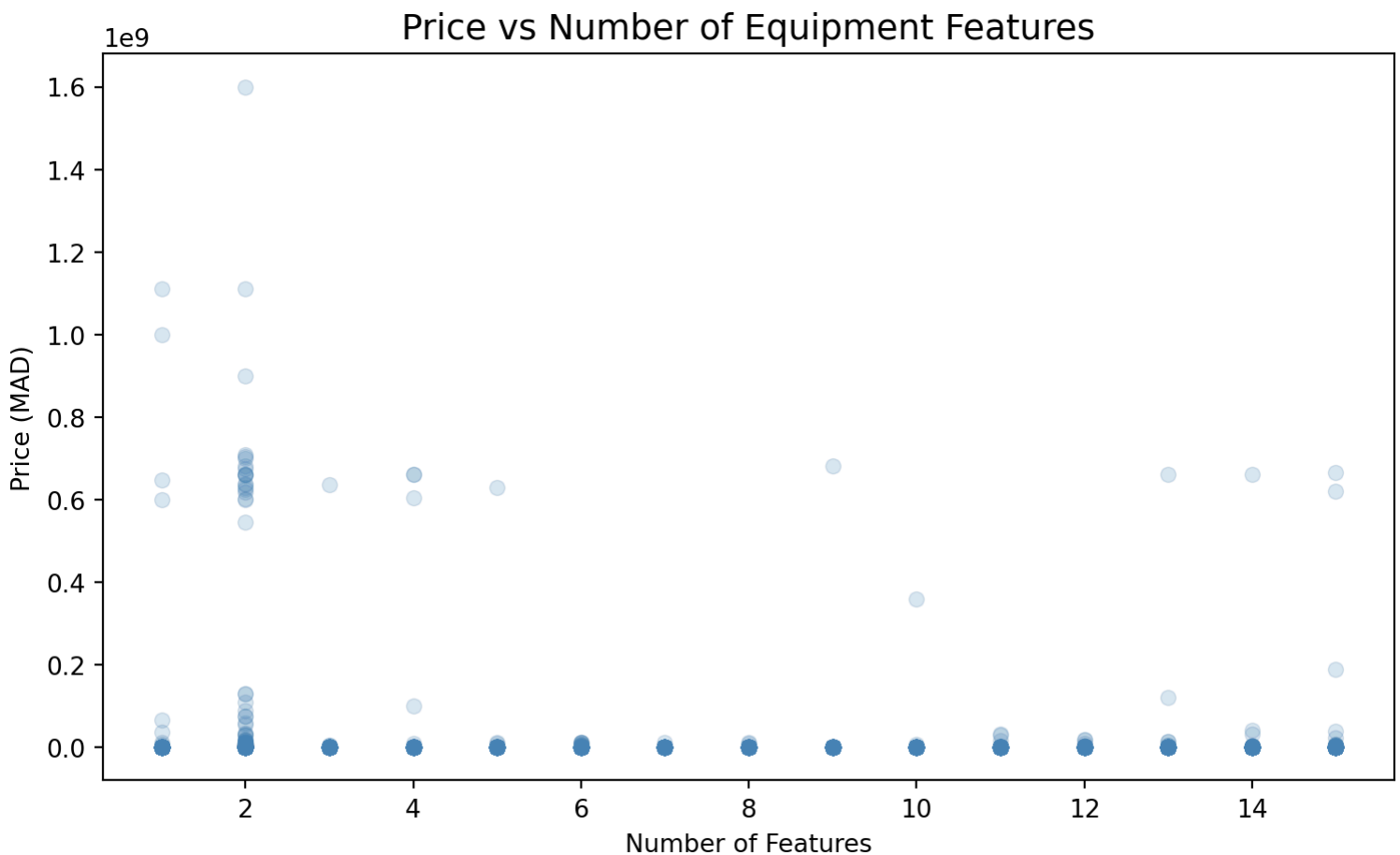


Answer: Safety standards rule! **ABS, Airbags, Power Steering, and Air Conditioning** are by far the most listed

Q15: Do more features = higher price?

▶ Code

```
Correlation (# of features vs price): -0.009
```

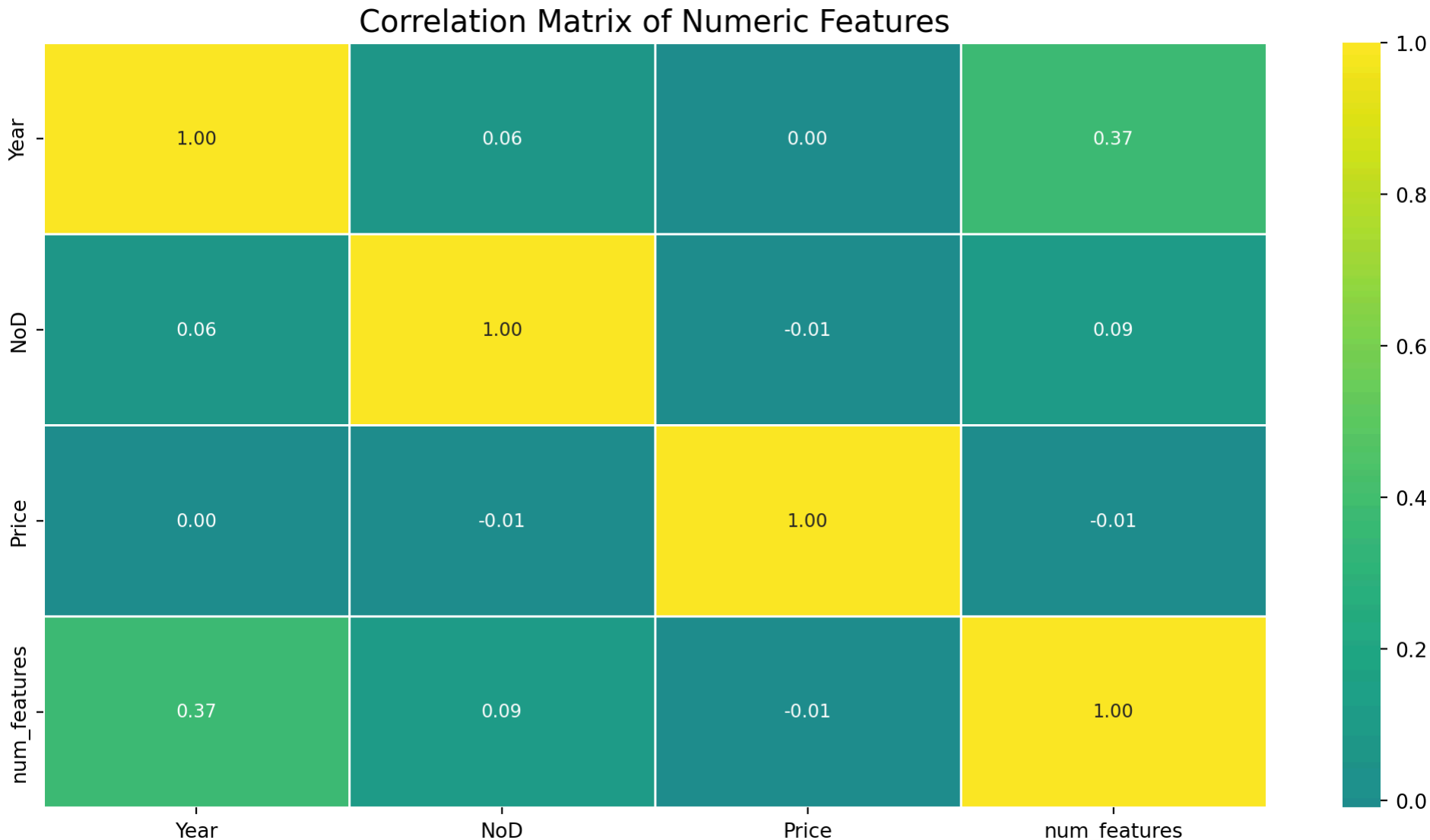


Answer: A **positive correlation** exists! More equipped cars do tend to fetch higher prices.

Part 6: The Big Picture

Q16: What actually correlates with price?

- Code



Answer: `Year` and `Fiscal Power` show the strongest correlations with `Price`. Higher year (newer) and more

Saving the Cleaned Data

Let’s save our squeaky clean dataset before moving on to the next chapter!

▶ Code

```
Dataset successfully saved as 'cleaned_data.csv'!
```

The Story So Far…

We've uncovered the key drivers of used car prices:

Factor	Effect on Price
Brand	Porsche > Dacia (a lot!)
Year	Newer = More expensive
Condition	Excellent >> Fair
Gearbox	Automatic > Manual
First Owner	Significant premium
Equipment Count	More features = higher price
Fiscal Power / Year	Strongest numeric correlators

What’s Next?

With a **clean**, **well-understood** dataset in hand, the adventure continues:

- **PCA & Dimensionality Reduction** — Compress features while keeping the signal
- **Machine Learning Model Training** — Regression to predict `Price`
- **Model Evaluation** — Which algorithm wins? How accurate can we get?

Thanks for your attention!

BAHLAOUI Ahmed · ENNACKHLAOUI AYA · LOGRAINE Wiam

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